



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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18th August, 2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

Executive Summary

Summary of Methodologies

1. Data Collection:

Via SpaceX API and web scraping records from Wikipedia.

2. Data Cleaning & Preparation:

After cleaning and formatting, data were stored into SQLite database to perform required SQL queries.

3. Exploratory Data Analysis.

4. Feature Engineering and standardizing the data.

5. Interactive Visualizations:

Display of launch sites and success rates on maps using Folium.

6. Model Building & Evaluation:

- Set up SVM, Decision Trees, and K-Nearest Neighbors models.
- Tuning hyperparameters with GridSearchCV.
- Model evaluations using test data accuracy.

Summary of Results

1. Data Insights:

- Identified factors influencing Falcon 9 first stage landings.
- Visualized geographical patterns and success rates.

2. Model Performance (Accuracy):

Best model : Decision Tree with 88.9% Accuracy.

Other models : SVM & K-Nearest Neighbor reached 83.3% Accuracy

3. Key Findings:

Launch Site and Payload Mass seem to impact the success of a landing

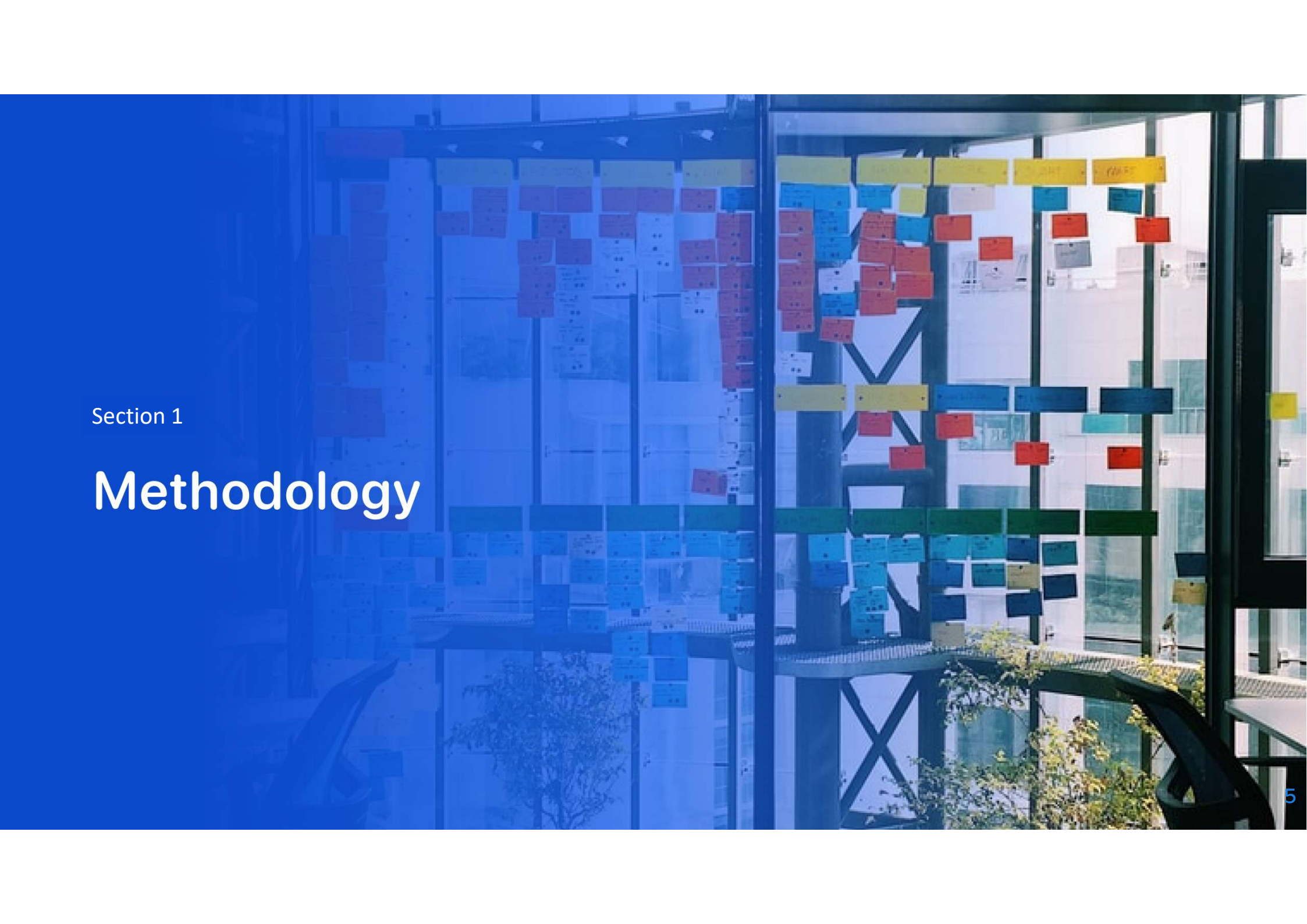
Introduction

Background

In this capstone project, we aim to predict the successful landing of the Falcon 9 rocket first stage. SpaceX advertises rocket launches at a significantly lower cost compared to other providers, largely due to their ability to reuse the first stage of the rocket. By accurately predicting landing success, we can estimate launch costs and provide valuable insights for companies bidding against SpaceX.

Questions to answer

- The factors that influence the success of the Falcon 9 first stage landing
- The prediction of the landing outcome using machine learning models
- Identify the best machine learning model in terms of landing outcome predictions



Section 1

Methodology

Methodology

Executive Summary

- **Data collection methodology:**
 - Data on Falcon 9 launches were collected using the SpaceX API : launch dates, sites, payloads, and outcomes and Web Scraping from Wikipedia webpage
- **Perform data wrangling**
 - Here we mainly handled variables with missing values, standardized the formats, created the target variable for the machine learning step (there are several possible values for good and bad outcomes, we grouped into 2 cases 0 = bad outcome, 1 = good outcome) and performed one-hot-encoding for categorical variables.
- **Perform exploratory data analysis (EDA) using visualization and SQL**
 - We used matplotlib and seaborn libraries to visualize success rates, payloads, and launch sites
 - SQL queries were used to answer specific questions

Methodology

Executive Summary

- **Perform Interactive Visual Analytics Using Folium and Plotly Dash:**
 - Folium was used to create interactive maps displaying launch sites and outcomes.
 - A Plotly Dash application was developed to create interactive visualizations with components like dropdowns and sliders. The Dash app allows the visualization and analysis of launch success rates for different payload ranges.
- **Perform Predictive Analysis Using Classification Models**
 - Built and evaluated different classification models including Logistic Regression, SVM, KNN, and Decision Trees.
 - The GridSearchCV function was used for hyperparameter tuning.
 - The models were evaluated based on the accuracy metric

Github : <https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>

Data Collection

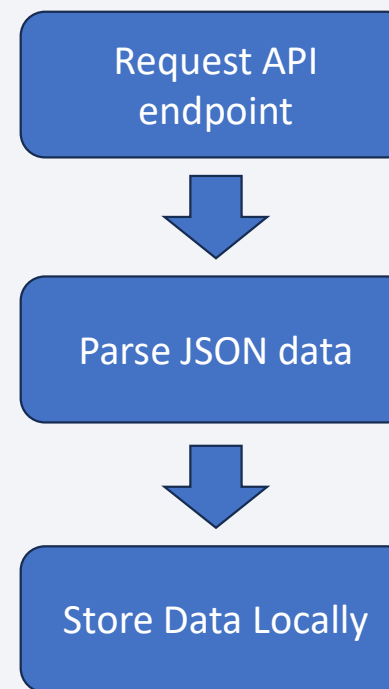
- Step 1 : SpaceX API
 - API request > Data fetching > Local data storage
- Step 2 : Web Scraping Wikipedia webpage
 - Extract HTML table > Data parsing with BeautifulSoup > Create a dataframe
- Step 3 : Datasets combinations
 - Merging SpaceX API data and Wikipedia data > create the final dataset

Data Collection – SpaceX API

- Step 1: Initiate API Request
- Step 2: Parse API Response
(Convert API response from JSON to a Python dictionary) and extract relevant fields
- Step 3: Store Data Locally

GitHub:

<https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>

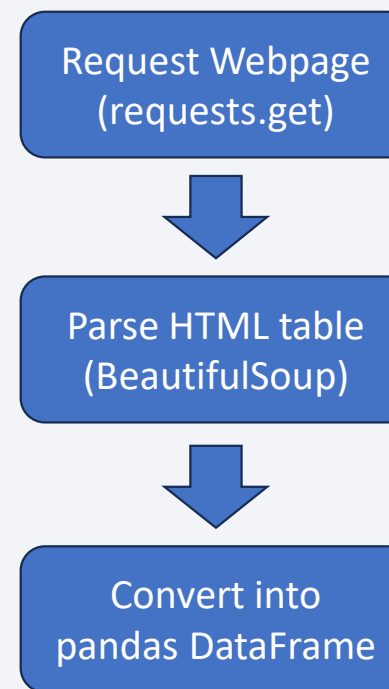


Data Collection – Scraping

- Step 1: Initiate Web Scraping (requests library to fetch the HTML content of the Wikipedia page).
- Step 2: Parse HTML Content : Use BeautifulSoup to Extract the HTML table containing Falcon 9 launch records.
- Step 3: Convert to a pandas DataFrame

Github :

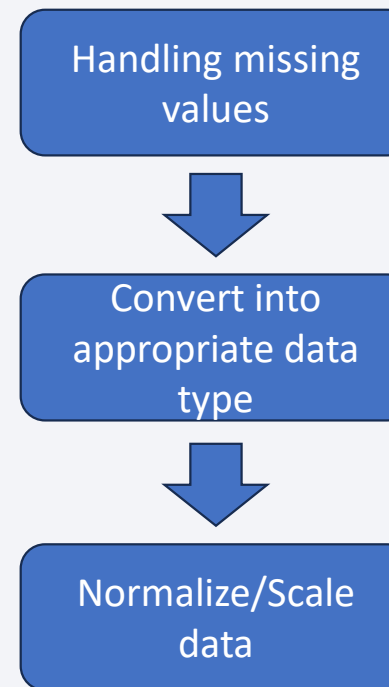
<https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>



Data Wrangling

- Collected data were :
 - Cleaned by handling missing values (by imputing them or by removing columns with too many missing values)
 - Transformed by converting the features into appropriate data types (e.g. dates, numeric fields...)
 - Normalized by scaling numerical features
 - New features were created (e.g. year by deriving from date) and one-hot-encoding was performed on categorical data to make the dataset suited for the machine learning part.

GitHub :
<https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>



EDA with Data Visualization

- Several visualizations were performed in the analysis :
 - Histograms for understanding the distribution of numerical features
 - Bar charts to compare categorical features in frequencies
 - Line charts to spot any trend over time
 - Scatter plots to explore relationship between 2 numerical features
 - Heatmap for analyzing correlations between features
 - Boxplot for feature distribution and outliers detection

GitHub: <https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>

EDA with SQL

- Among SQL queries performed for data analysis, we had :
 - Aggregation queries (e.g. calculation of success/failure rates)
 - Join queries for combining information from different tables (add rocket data to launches data)
 - Filtering queries to only access data of interest (e.g. focus on specific outcomes failure vs success)
 - Sorting queries to order data by date
 - Subqueries for more complex calculations (average payload mass per launch site)

GitHub: <https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>

Build an Interactive Map with Folium

- Map Objects :
 - Markers (Launch sites) : To pinpoint exact launch locations for spatial reference
 - Circles (Proximity zones) : To provide a visual representation of safety perimeters
 - Lines (Connections) : to highlight relationships between launch sites

GitHub : <https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>

Build a Dashboard with Plotly Dash

- Success Pie Chart (distribution of successes and failures)
 - Interaction : Launch sites selection with dropdown menu
- Success vs Payload scatterplot
 - Interaction : Range slider to adjust payload mass interactively

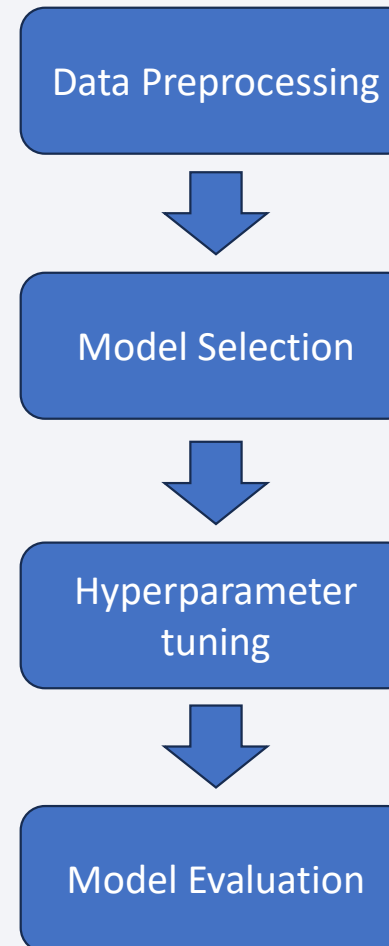
GitHub : <https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>

Predictive Analysis (Classification)

- Step 1 : Data preprocessing (scaling + train/test split)
- Step 2 : Model selection (KNN, Logistic Regression, SVM and Decision Tree)
- Step 3 : Hyperparameter tuning
- Step 4 : Model evaluation
- Step 5 : Selection of the “best” model (Accuracy metric)

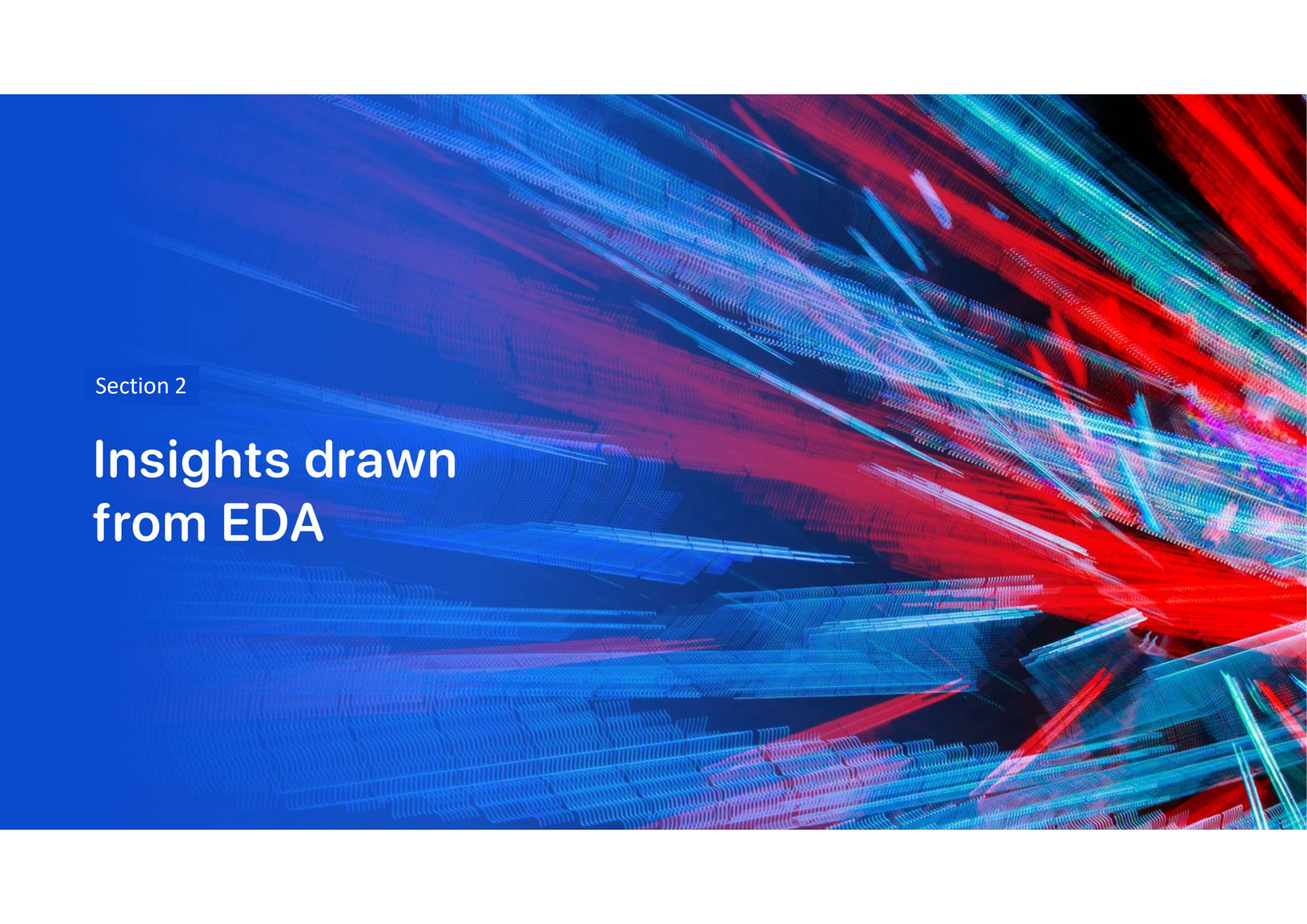
GitHub :

<https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>



Results

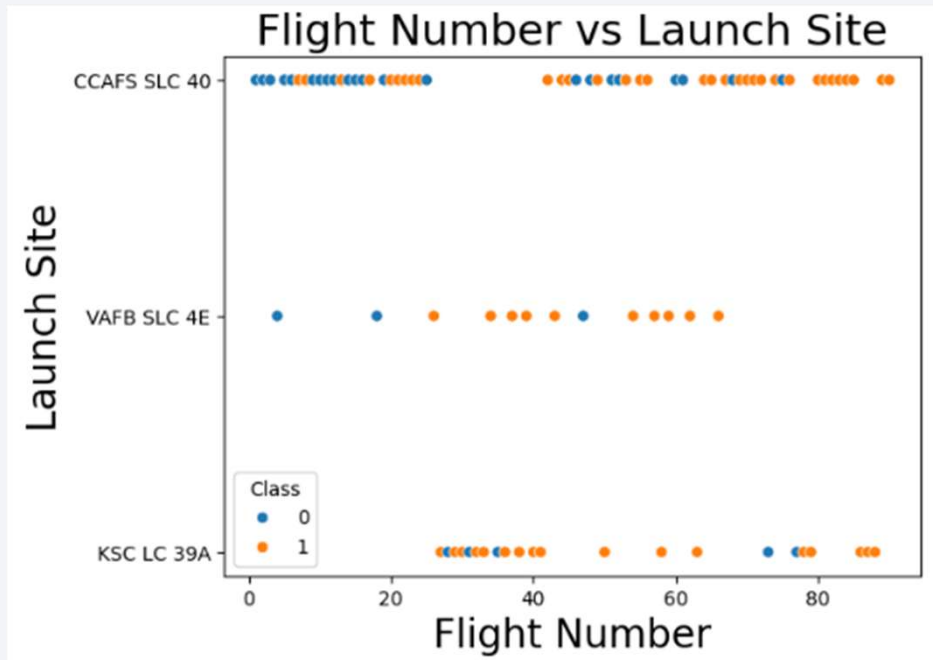
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



Section 2

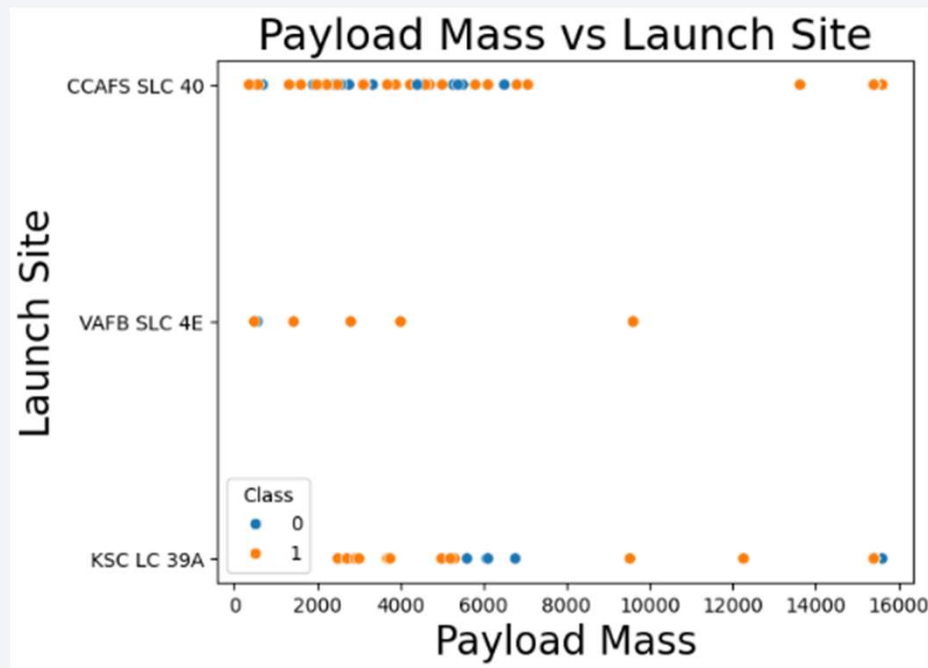
Insights drawn from EDA

Flight Number vs. Launch Site



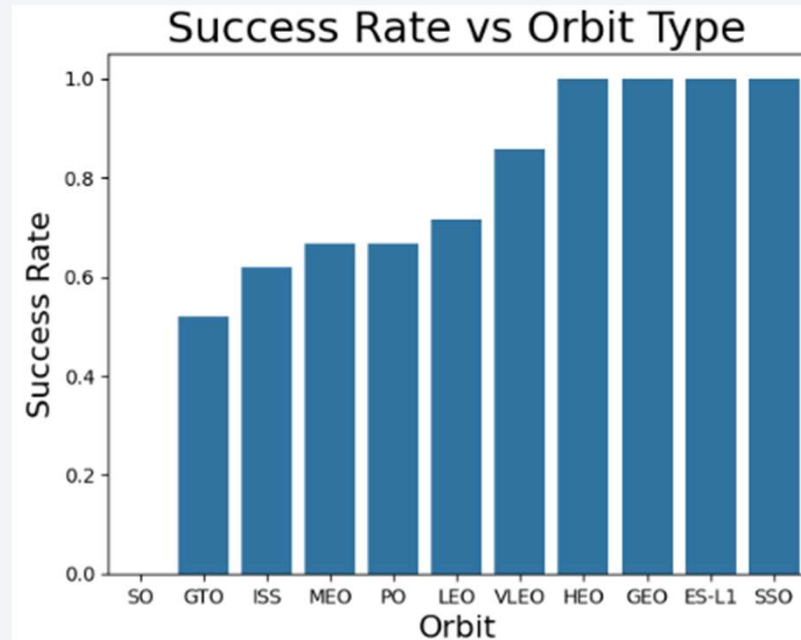
- Each site presents a mix of successful and failed outcomes.
- Fewer launches from the VAFB SLC 4E with however a better success rate.
- Based on Flight Number, we can say that CCAFS site exhibits constant activity with a gap corresponding to more launches done from the 2 other sites. Launches from KSC LC started later.

Payload vs. Launch Site



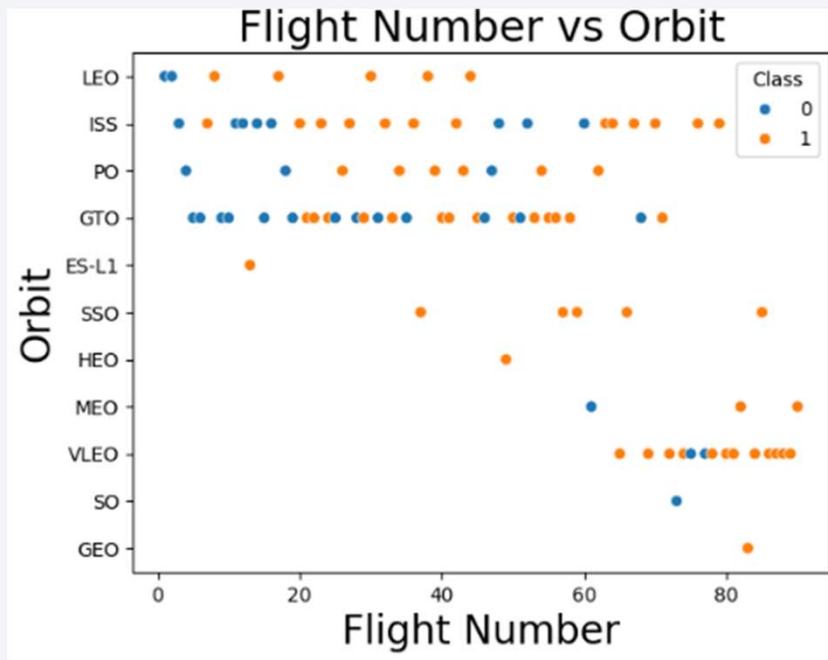
- Most launches from the CCAFS SLC 40 and VAFB SLC 4E sites handle payloads below 10,000 kg, while the KSC LC 39A sites has a wider range of payload masses, suggesting that goals of these missions may be different.
- KSC LC 39A site has a higher proportion of high-capacity launches suggesting that it may be more suited for this kind of mission.

Success Rate vs. Orbit Type



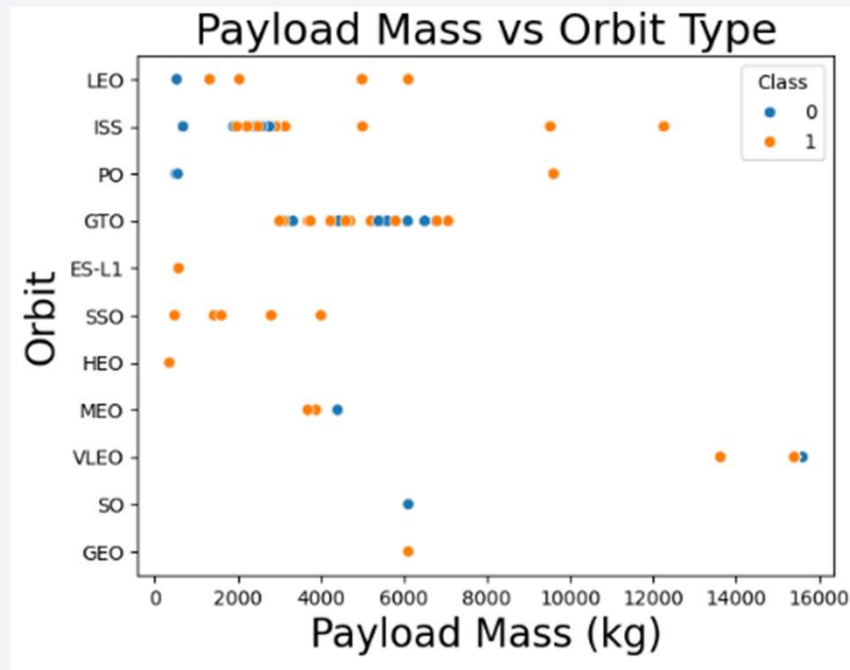
- Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits have higher success rate (perfect scores) indicating these orbits are highly predictive for successful missions
- Missions to GTO orbit has 1 chance over 2 to succeed indicating high challenges for this kind of orbit.

Flight Number vs. Orbit Type



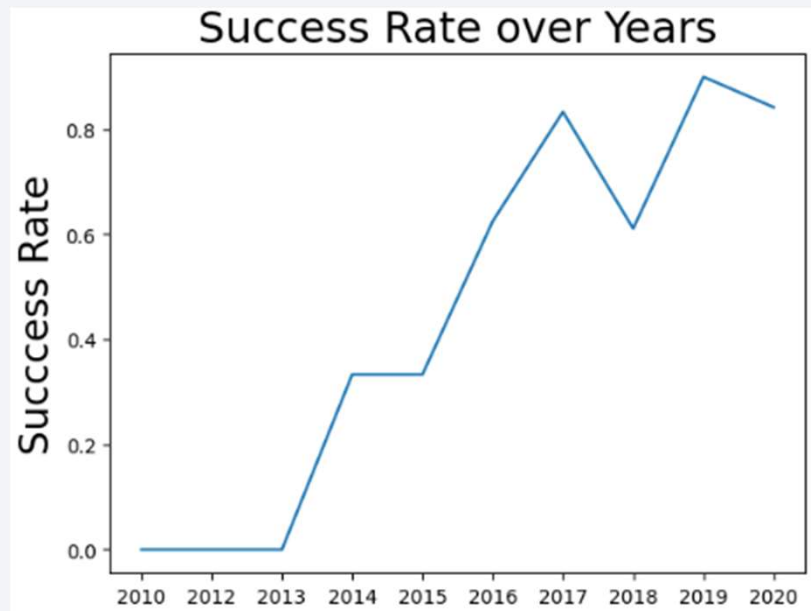
- Missions to ISS cover almost the whole range of Flight Number indicating that this kind of mission represent an important goal for SpaceX.
- VLEO missions seem to be more recent Flight Number and exhibit a higher success rate suggesting a shift in the strategy of SpaceX

Payload vs. Orbit Type



- Launch to VLEO orbit show higher capacity in terms of payload mass.
- Some of the launches to ISS orbit has a payload mass higher than 8000 kg.
- The other orbits seem more dedicated to missions with lower capacity.

Launch Success Yearly Trend



- The success rate trend tends to be constantly increasing over the years.
- After a stagnation around 40% of success in 2014 and 2015, the success rate reaches 50% in the following year.
- We however observed a dip in 2018

All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[7]: %sql select distinct Launch_Site from SPACEXTABLE
```

```
* sqlite:///my_data1.db  
Done.
```

```
[7]:
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- This SQL query displays the names of the different launch sites. The keyword “distinct” is the one that allows the selection of unique names.

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[18]: %sql select * from SPACEXTABLE where Launch_Site like 'CCA%' limit 5
* sqlite:///my_data1.db
Done.
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- To find the first records with launch site starting with 'CCA', the keyword "like" is used in conjunction with the "%" wildcard character.
- The "limit" keyword allows us to filter for the first 5 records.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[10]: %sql select sum(PAYLOAD_MASS__KG_) as total_payload_mass from SPACEXTABLE where Customer=='NASA (CRS)'  
* sqlite:///my_data1.db  
Done.  
[10]: total_payload_mass  
      45596
```

- We sum the payload mass over all the entries where the customer is “NASA (CRS)”. The “Where” clause allowed the filtering step.
- We aliased the name of the result to get a meaningful variable name

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[11]: %sql select avg(PAYLOAD_MASS_KG_) as average_payload_mass from SPACEXTABLE where Booster_Version like 'F9 v1.0%'
* sqlite:///my_data1.db
Done.
[11]: average_payload_mass
      340.4
```

- As for the previous query, the “where” clause allowed the filtering for the Booster version.
- Aliasing the result name gives a clearer variable name.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[12]: %sql select min(Date) as first_success_ground_pad from SPACEXTABLE where Landing_Outcome == 'Success (ground pad)'  
* sqlite:///my_data1.db  
Done.  
[12]: first_success_ground_pad  
2015-12-22
```

- Combination of where clause and the use of “min” function on the date field allowed us to achieve the expected result : the date of the 1st successful landing on ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[13]: %sql select Booster_Version, PAYLOAD_MASS_KG_ from SPACEXTABLE where Landing_Outcome == 'Success (drone ship)' and PAYLOAD_MASS_KG_ between 4000 and 6000
* sqlite:///my_data1.db
Done.
```

```
[13]:
```

Booster_Version	PAYLOAD_MASS_KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

- We used 2 filtering clauses :
 - Landing outcome to get only the records with Success (drone ship)
 - Payload mass to get the ones between 4000 and 6000 kg

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[14]: %sql select Mission_Outcome, count(*) as total from SPACEXTABLE group by Mission_Outcome
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[14]:
```

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- We listed all kind of successes and failures
- We observed only 1 failure in flight

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
[19]: %sql select Booster_Version from SPACEXTABLE where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE)
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[19]: Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

- A subquery is used to filter the data. This subquery selects the maximum payload mass from the table.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[24]: %sql
SELECT
  CASE SUBSTR(Date, 6, 2)
    WHEN '01' THEN 'January'
    WHEN '02' THEN 'February'
    WHEN '03' THEN 'March'
    WHEN '04' THEN 'April'
    WHEN '05' THEN 'May'
    WHEN '06' THEN 'June'
    WHEN '07' THEN 'July'
    WHEN '08' THEN 'August'
    WHEN '09' THEN 'September'
    WHEN '10' THEN 'October'
    WHEN '11' THEN 'November'
    WHEN '12' THEN 'December'
    ELSE 'Unknown'
  END AS month,
  Booster_Version,
  Landing_Outcome,
  Launch_site
FROM SPACEXTABLE
WHERE SUBSTR(Date, 1, 4) = '2015'
  AND Landing_Outcome = 'Failure (drone ship)';
```

```
* sqlite:///my_data1.db
Done.
```

```
[24]:
```

month	Booster_Version	Landing_Outcome	Launch_Site
January	F9 v1.1 B1012	Failure (drone ship)	CCAFS LC-40
April	F9 v1.1 B1015	Failure (drone ship)	CCAFS LC-40

- The “CASE WHEN” statement allows to get the month names instead of just a number.
- We then filtered to get the failures for landing outcomes in drone ship for the year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[21]: %sql select count(*) as cnt, Landing_Outcome from SPACEXTABLE where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by cnt desc
```

Done.

```
[21]: cnt    Landing_Outcome
-----
10      No attempt
5       Success (drone ship)
5       Failure (drone ship)
3       Success (ground pad)
3       Controlled (ocean)
2       Uncontrolled (ocean)
2       Failure (parachute)
1       Precluded (drone ship)
```

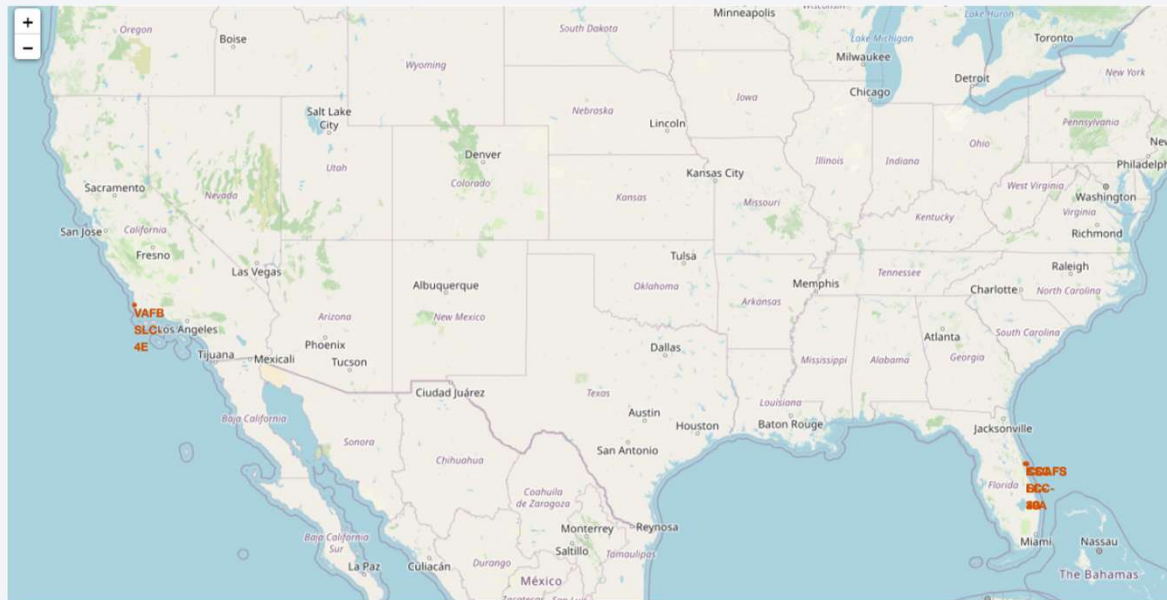
- We aggregated the data by landing outcomes and count the occurrences for each landing outcome with the “COUNT” function after filtering for the launches between 2010-06-04 and 2017-03-20.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue gradient on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

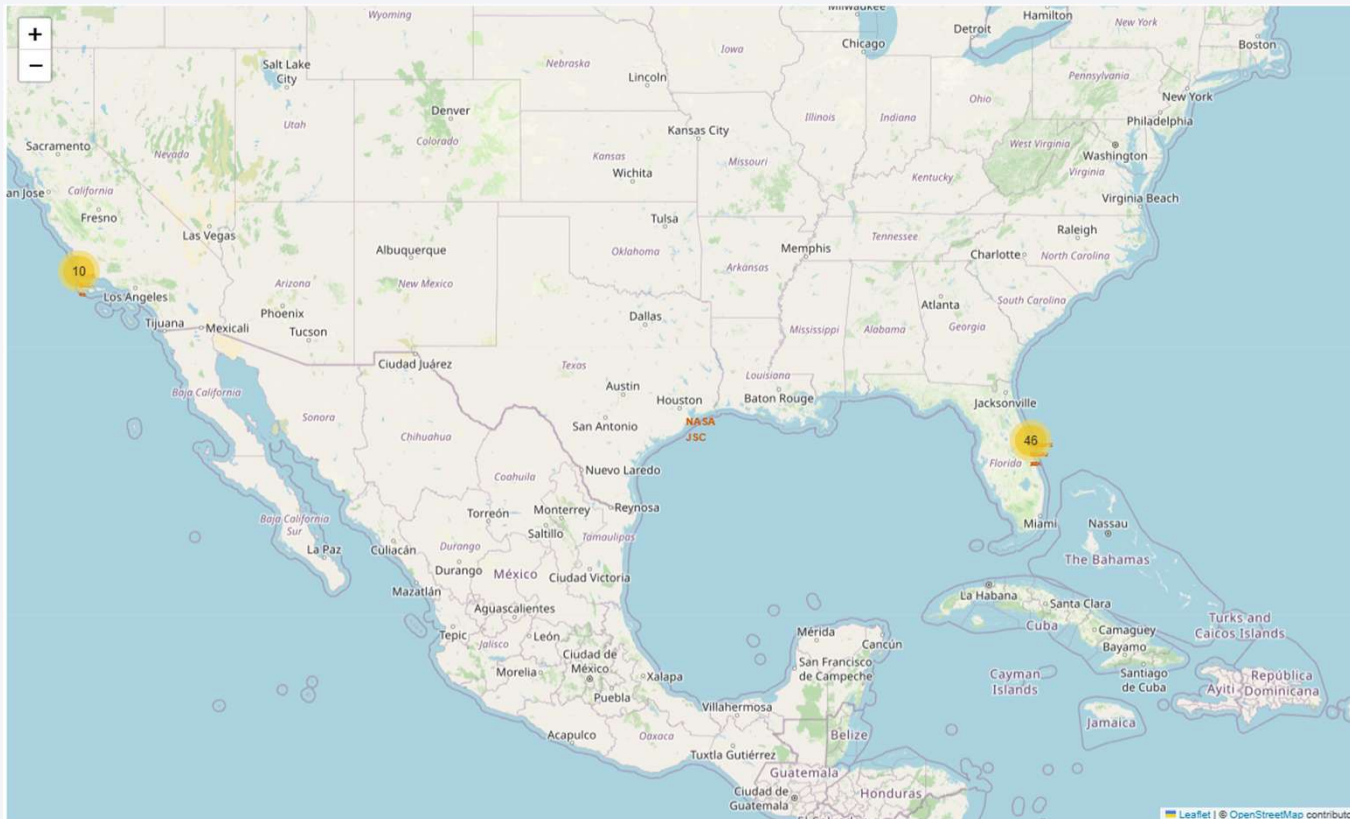
Launch Sites Proximities Analysis

Task 1 : Mark Launch Sites on the Map



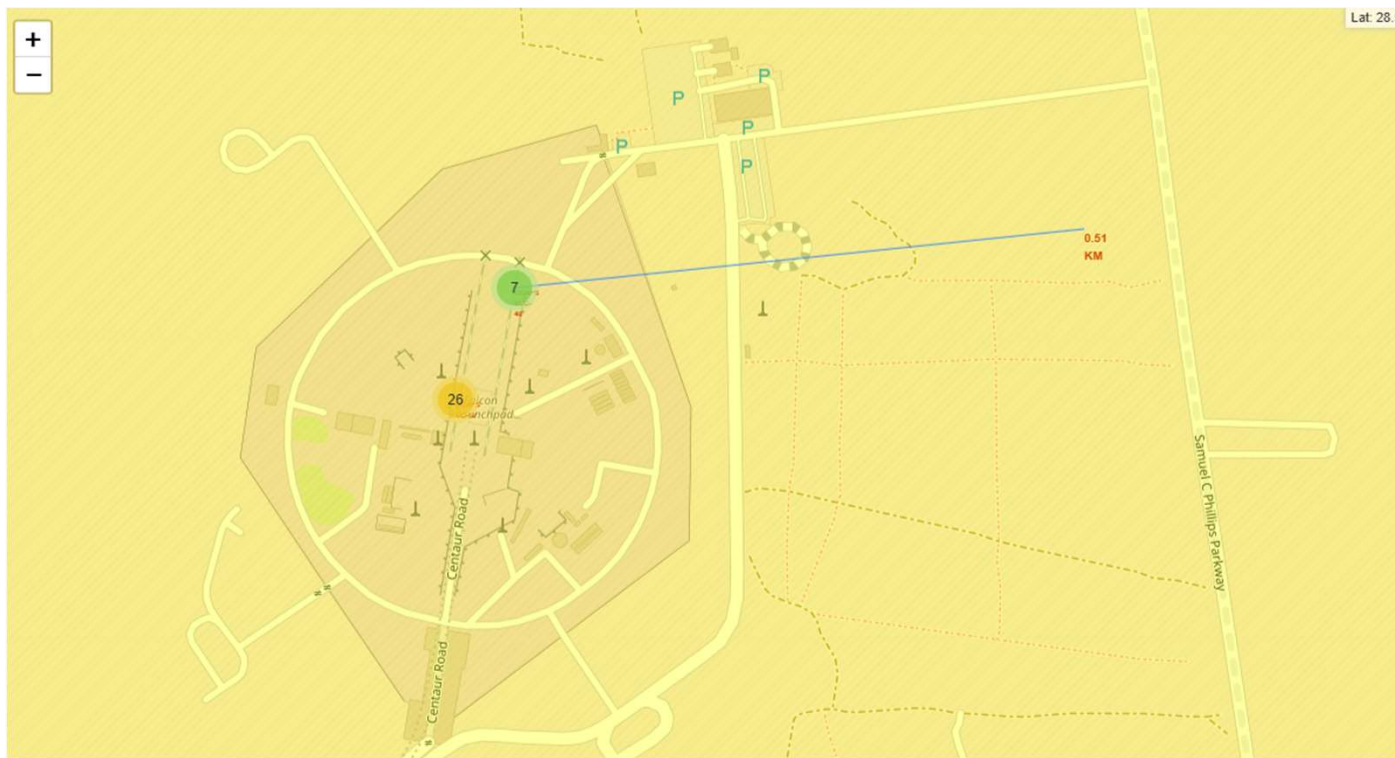
- All Launch Sites are coastal.
- Either on the West Coast (VAFB SLC-4E) or on East Coast (CCAFS LC-40, CCAFS SLC-40 and KSC LC-39A)

Task 2 : Mark Successful/Failed launches

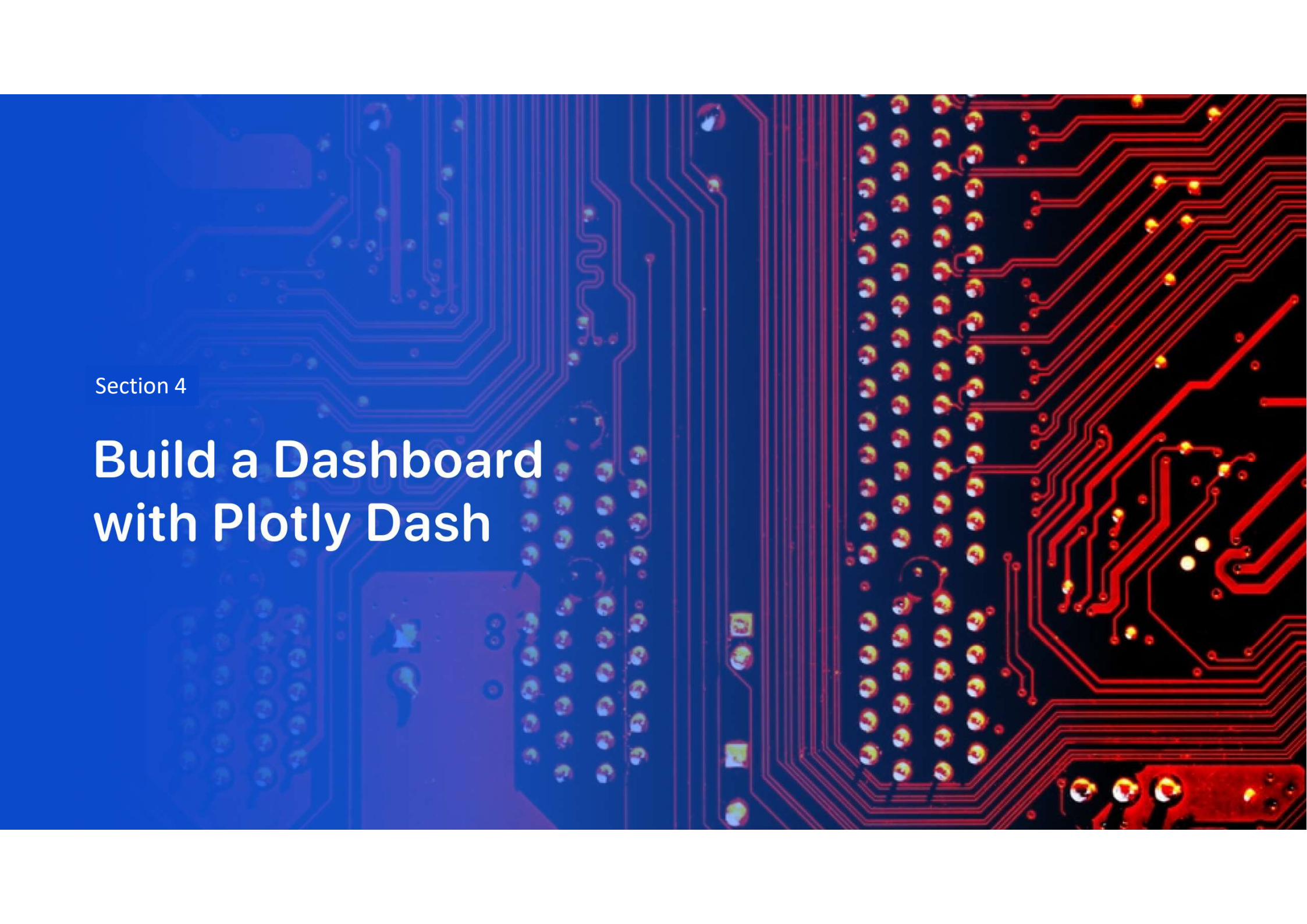


- The circles represent the area of all launches
- The numbers represent the success on each Launch Sites

Task 3 : Calculate Distance Launch Sites - Proximities



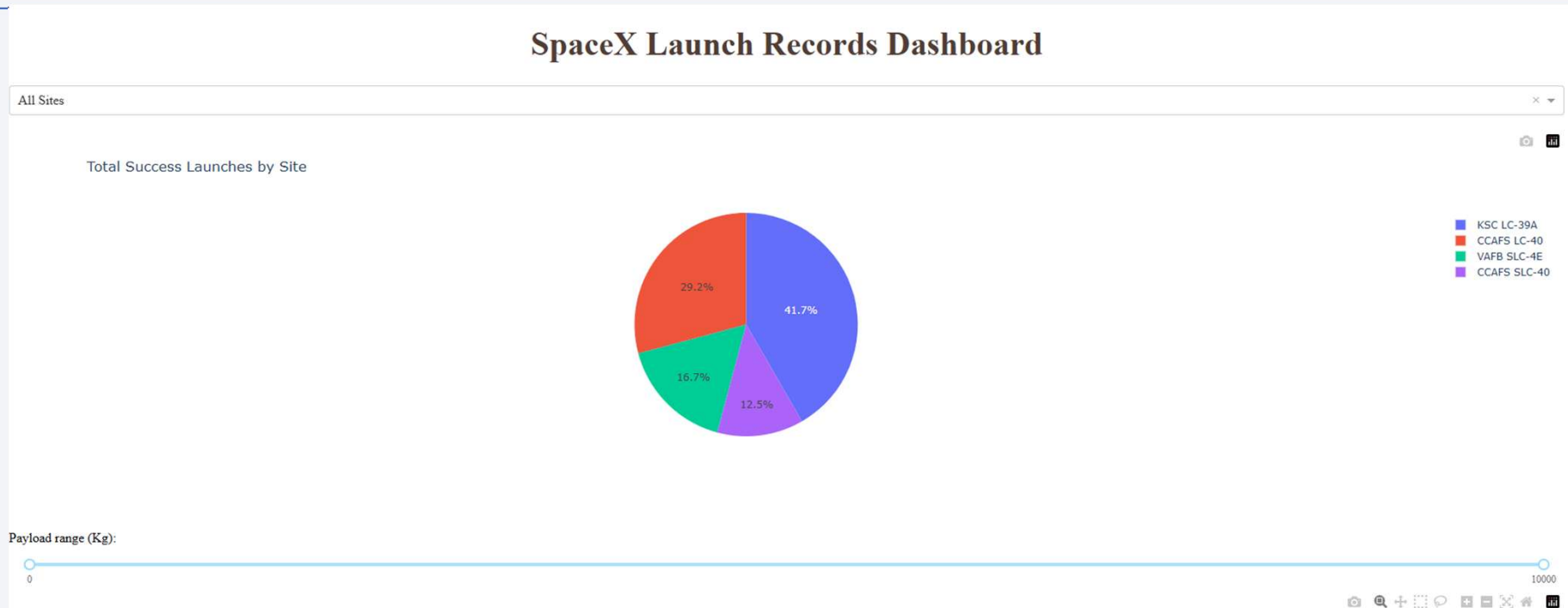
The distance between the CCAFS SLC-40 launch site and the closest coastline is around 0.51 kilometers. Close proximity to the coastline facilitates over-water flight paths and recovery while ensuring minimal risk to the population in the neighborhoods.



Section 4

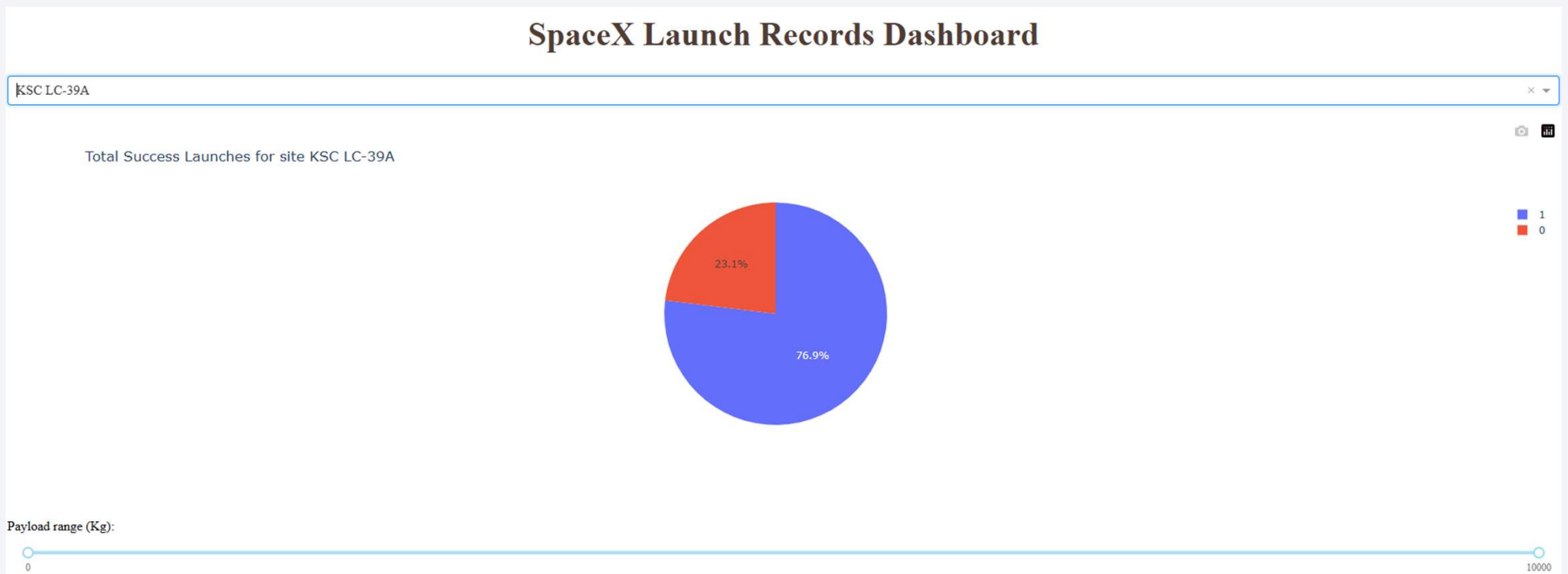
Build a Dashboard with Plotly Dash

Success Rate by Launch Site



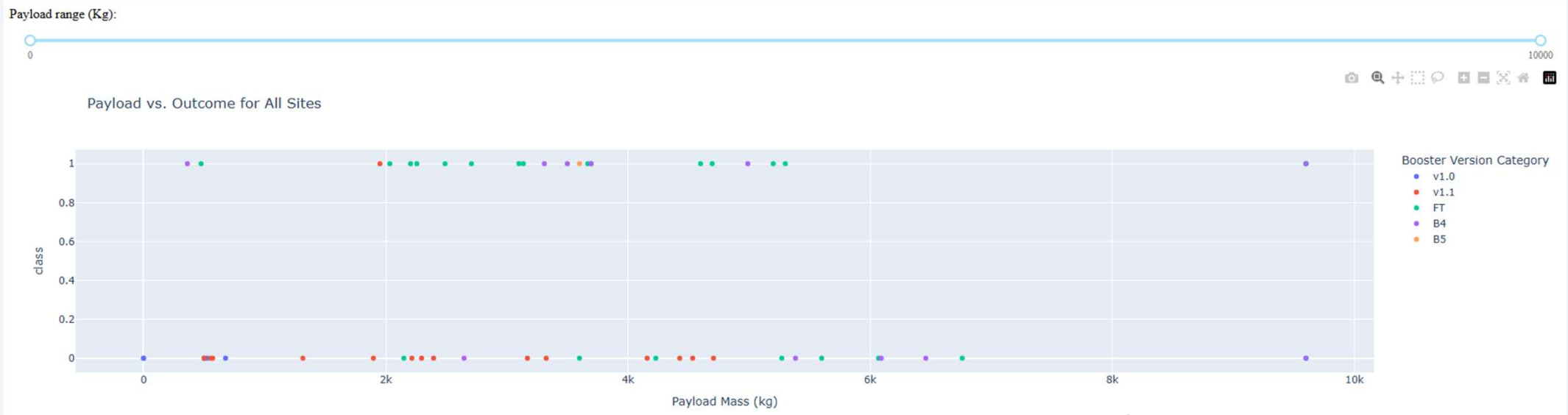
- The KSC LC-39A is the site with the highest number of successful launches. It represents almost the half (~42%) of all successful launches.

The KSC LC-39A Launch Site



- More than $\frac{3}{4}$ of launches (76.9%) from the KSC LC-39A site were successful.
- The other sites exhibit poorer results (with less than 50% of successful launches)

Impact of Payload Mass on Outcome



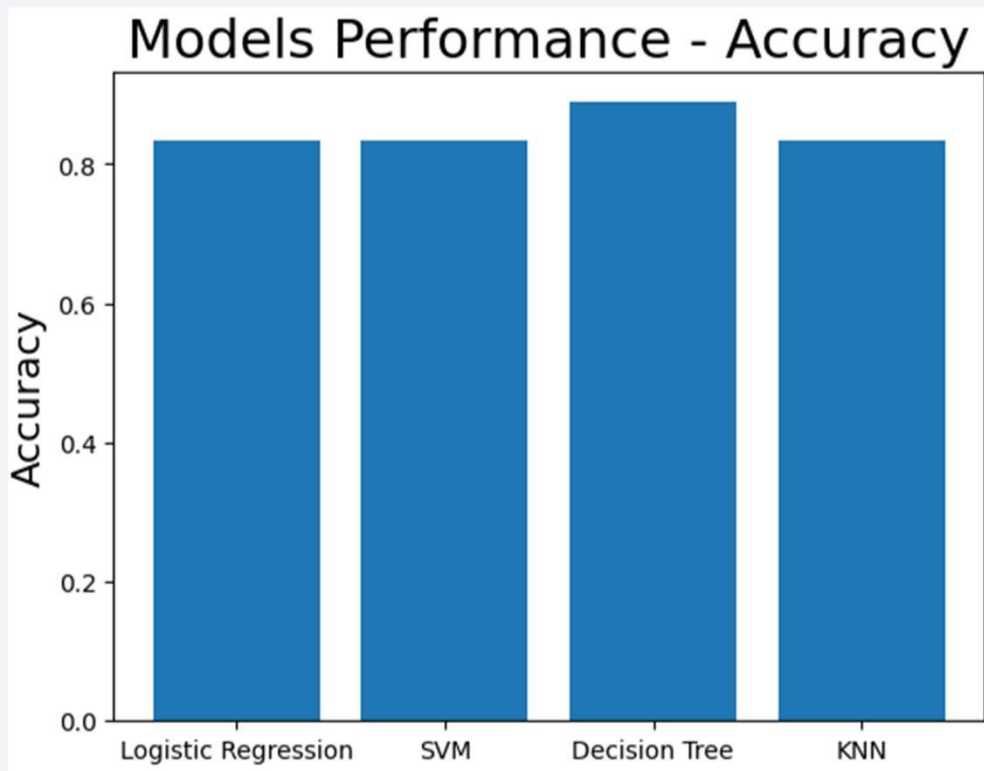
- The launches with FT booster version represent the majority of successes (class 1 outcome).
- The v1.0 version was clearly less used and may require more tests to evaluate its true performance.
- The v1.1 booster version had a higher proportion of failures (class 0 outcome)
- Most of the successful launches had a narrower range of payload masses.



Section 5

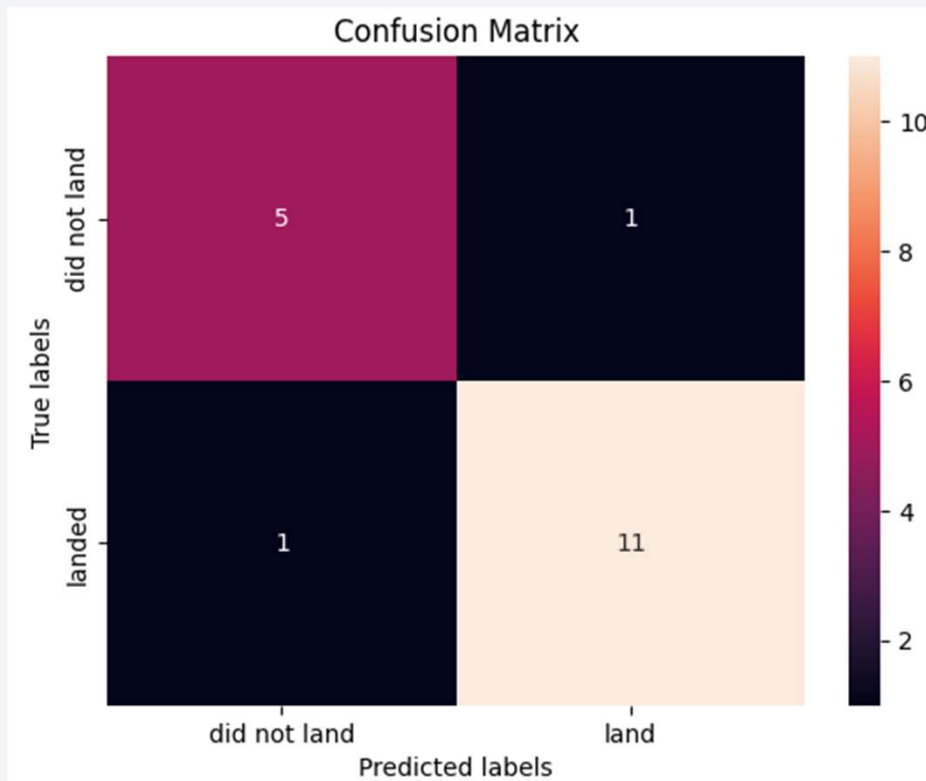
Predictive Analysis (Classification)

Classification Accuracy



- Decision Tree achieved the best accuracy score on test data (0.89).
- The other models achieved similar accuracy score (0.83)

Confusion Matrix – Decision Tree



- The **top left** quadrant represents the number of **True Negative** : negative outcomes correctly predicted as negative by the model.
- The **top right** corresponds to the **False Positive** : classified as positive outcome (“land”) while the true outcome was negative (“didn’t land”)
- The **bottom right** quadrant represents the **True Positive** : positive outcomes correctly predicted as positive by the model.
- The **bottom left** corresponds to the **False Negative** : classified as negative outcome (“didn’t land”) while the true outcome was positive (“land”)

Conclusions

- "CCAFS LC-40" launch site represents the most contributor to successful landing (41.7% of all successful landings). This site may be particularly suited for future missions.
- The "FT" booster has a high success rate indicating high reliability. It should be used in the next missions.
- No strong relationship between Payload Mass and Success Rate

Additional analyses may be needed to draw definitive conclusion, but we can already say that Booster version and Launch Site may play a role in the success of a mission.

Thank you!



Appendix

- All material (notebooks with code and figures) is provided in the following GitHub repository :

<https://github.com/FouadEddahri/Applied-Data-Science-Capstone.git>